CS-80 Reference: plan completion and handover

What the improvement plan delivered, what is still open, and exactly how the next agent should finish it.

Repository `cheshmakzanoon-ops/yamaha-cs80` · state of `main` at `6cc9b13` (README PR #4), code at `55bb043` · 31 October 2026 · licence AGPLv3

At a glance.

- All eight plan workstreams (Phases 1–6) are merged to `main` through PR #2, with CI fixes in PR #3. Phase 7 packaging is automated and the project is licensed AGPLv3.
- CI on `main` at `55bb043` passed everywhere: four-platform build and tests, pluginval strictness 10 on every platform, the release installers, the DSP, compatibility and sanitizer jobs, and the screenshot render.
- The README on `main` describes the current emulation, links the new builds and installers, and shows freshly rendered screenshots.
- **Not finished:** the CPU targets, one retrigger-click outlier, the cause of the intermittent Linux pluginval exit 139, and a few partly implemented items (listed in section 3). Signing, DAW testing and a trademark review need the owner.

1. Scorecard against the plan's targets

Metric	Target	Before the plan	Now	Met?
Worst saw/pulse aliasing, circuit engine (ring off)	≤ −70 dB	−33.8 dB	−88.3 dB (48 kHz)	**yes**
Worst tone-tab/program peak at default volume	≤ 0 dBFS	+3.9 dBFS	−0.24 dBFS	**yes**
Worst voice-steal click score	≤ 1	1.17	0.92	**yes**
Worst retrigger click score	≤ 1	—	1.51 ("Bass Stack"); 281 of 283 rows ≤ 1	**no**
CPU, 8 voices, 48 kHz, 256-frame blocks	0.15–0.20 × realtime	0.27–0.30	0.51–0.61 median, p99 up to 1.00	**no**
Idle CPU	≤ 2 %	6–7 %	11 %	**no**
Legacy audio	byte-identical	—	byte-identical to `d56911a`	**yes**

The measurements come from `cs80_engine_audit --suite all --rates 44100,48000,96000` and `cs80_source_factory_benchmark` on a shared 4-core container. They are software measurements, not comparisons with a hardware CS-80.

2. What has been done

Plan step	What was implemented	State
Phase 0 foundations	Coefficient caching; state schema (now version 3); `docs/STATUS.md` as the single status page; smoothing (moved into Phase 1).	**done**

1.1 / 1.1b IG00156 filter	Nodal stage (`Ig00156Filter.h`): input networks, 1.2 MΩ/180 pF branch, 1 µF couplings, in-loop OTA saturation, Newton solve, declared Q-block law and cutoff calibration. Truth model in `reference/IG00156Network.h`.	**done, 2 misses**
1.2 Oscillator	Reconstructed event-driven oscillator promoted to the circuit engine. Its residual table is shared across instances, and a piecewise sine shaper replaces the 3.5 % third-harmonic placeholder. Aliasing is now −88 dB at 2×.	**done**
1.3 Envelopes	Single-RC law, time-to-−60 dB, overshoot-target attack, filter-EG continuity on voice reuse.	**done**
1.4 VCA (IG00151)	Separate linear and exponential inputs, the LI node, a −52 dB AC-coupled pedestal, constant-power pan, a model doc and mutation checks.	**done; 1 click outlier**
1.5 Gain staging	−6 dB output reference and an inferred driver ceiling.	**done**
1.6 Oversampling	Eco 2× / Standard 2× / HQ 8×. Offline bounces use HQ. Each mode has its own decimator, and latency is reported correctly.	**done**
1.7 Engine default flip	New instances use the circuit engine; old sessions load as Legacy. The two old toggles are now ENGINE (Circuit / Legacy) and PURE CS-80.	**done**
Smoothing	5 ms sample-rate-independent smoothing on every continuous control (`Smoothing.h`).	**done**
2.1 Free-running voices	All 16 chains run continuously, with a cheap idle path and exact catch-up when a voice wakes.	**done**
2.2 KAS scan / sample-hold	Declared scan timing, droop, NAND(TR,SUS) hold and ninth-key reconsideration.	**done**
2.3 Keyboard contour	Declared 61-key map driving `KeyboardContour.h`. The C2/KAS6 conflict is reported openly.	**done**
2.4 Touch	TWS a/b/C nodes, the TRG5 diode-OR pressure bus, and velocity/pressure curves for different controllers.	**done**
2.5 Tolerances and drift	Declared 5 % resistor / 10 % capacitor draws mapped into all five trims, plus 1/f thermal drift with warm-up. The old ±22 cent sine remains as a "worn unit" option.	**done**
Phase 3 circuits	IG00150 SUB converter; ring modulator anchored to ADJ p19; OE1 tremolo; OE2 BBD with true L/R paths; PRA optical cell; EXP volume through the same cell; GENERAL/PHONES outputs; declared noise spectra.	**done**
4.1 Tone tabs	Per-cell review records and an opt-in refined BA/diode boundary. All 25 unknown cells remain labelled hypotheses.	**partial**
4.2 Memory panels	Electrical model and truth model, tested, but not yet driving audio.	**partial**
4.3–4.4 Library, MIDI	301 curated circuit-engine programs; a byte-identical 222-program Legacy Library; Program Change with bank select; ribbon CC.	**done**
5.1–5.7 UI/UX	Panel conformance audit; re-layout; embedded OFL typeface; zoom; editor split; toolbar; Studio view; voice monitor; relative faders, typed values and context menu; QWERTY play; tour and help; evidence view; golden images; hit-target and contrast tests.	**done**
Phase 6 modern layer	3 MSEGs, LFO waveforms, a 16-slot mod matrix, modern FX, Scala/KBM, MPE, voice modes, MIDI learn, undo/redo, A/B, randomise, host parameter groups. All of it is inert in PURE CS-80.	**done except 2 items**
Phase 7 release	`release.yml` builds a universal macOS `.pkg`, a Windows installer and a Linux `.deb`, and signs only when secrets exist. AGPLv3 `LICENSE`; qualification checklist.	**automated; owner items open**

3. What still needs to be done, and how

Items are ordered by value. For each one: the goal, where to work, the method, and when it counts as done.

3.1 CPU: reach 0.15–0.20 × realtime, idle ≤ 2 %

Why it misses. Profiling shows the reconstructed oscillator's event solver (about 29 % of instructions) and the nodal filters' Newton solve (about 22 %) dominate. Fast `exp2`/`pow` were no faster than glibc and were dropped; only the bounded-error `tan` was kept.

1. **Benchmark first.** Take a baseline with `build-core/cs80_source_factory_benchmark` (8 voices, 48 kHz, 256 frames, minimum of 3 rounds). Record the median, p99 and deadline misses.
2. **Oscillator.** In `src/core/ReconstructedOscillator.h`, compute the next branch event once per analytic segment and cache it, instead of solving every sample. Hold it to the existing oscillator tests and the alias suite.
3. **Nodal filter.** In `src/core/Ig00156Filter.h`, cache the Jacobian while the cutoff and Q inputs are unchanged. Skip the Newton iteration when the residual is already below tolerance.
4. **Engine split.** Split `Engine::processInternalSample` (`src/core/Engine.cpp`) into global, rank, mix and output parts. Store the 16 rank chains structure-of-arrays so the compiler can vectorise across ranks.
5. **Idle.** Find which always-on circuits cost about 0.08 × realtime with no notes held (chorus, noise, idle voices), and decimate them when idle.
6. **Done when** the targets are met, or the gap is documented with numbers. Legacy must stay byte-identical. Any change to circuit-engine audio needs one V2 re-pin (see section 4).

3.2 Retrigger click outlier ("Bass Stack", 1.51)

1. Reproduce it with `cs80_engine_audit --suite phrase` (which reports the steal, retrigger and note-off click scores) at 44.1 kHz. Legacy scores 0.63 on the same row.
2. Render the onset with `cs80_render` and compare the VCA level node, the AC-coupling state and the filter state at the retrigger instant (`src/core/Vca.h`, the free-running voice code in `Engine.cpp`).
3. Likely suspects: the pedestal coupling capacitor charging on a voice that was already free-running, or the envelope's overshoot target restarting from a non-zero level.
4. Fix the cause rather than adding a fade. Add a regression test for that program. Also look at the note-off scores above 1 (up to 1.65) on percussive programs.

3.3 Intermittent Linux pluginval exit 139

1. Code review and a valgrind teardown probe (`80cbe8c`) found no fault, so the cause is still open. Read `docs/VALIDATOR_DIAGNOSTICS.md` first.
2. Collect every failing CI log with the GitHub tools (`get_job_logs`) and note the seed, the phase (headless or GUI) and timing.
3. Check for timers or threads still alive when the plugin is unloaded. Candidates: the tour/help timers, the undo flush, `PaintCache`, static `juce::Image`/`Typeface` objects, and the embedded font.
4. Done only when a mechanism is proven and fixed. A passing rerun is not a fix.

3.4 Nodal filter accuracy

- **Top octave.** At 2× and 4× internal rate the small-signal gate fails for the 4 and 12 kHz cutoffs because of bilinear frequency warp. Pre-warp the integrator coefficients at the cutoff, or use a matched-z correction. Keep the tolerances in `tests/parity/tolerances-v2/` unchanged.
- **ADJ p11 HP setup point.** It reads 1.69 Vpp against the documented 4.3 ± 1 Vpp. Re-check the declared HP source resistance (75 kΩ) and the LP 150 kΩ reading, which the evidence ledger reads as a shunt. Record the result as evidence; do not tune to fit.

3.5 Unfinished features

Item	How to finish it
Sample-accurate automation	Split `processBlock` at host parameter-change timestamps (VST3 sample offsets) for the key parameters: cutoff, resonance, levels, pitch. Add a host test with a mid-block step.
MTS-ESP client	Vendor ODDSound's MIT-licensed client (`libMTSClient.h/.cpp`, with its licence) into `deps/mts-esp/Client/` or the source tree. Then compile-test the existing shim branch.
Memory-panel loading in audio	Wire `src/core/MemoryPanels.h` into the tone-selection path on the circuit engine only. It changes circuit-engine audio, so it needs a V2 re-pin.
25 tone-tab cells	Get native-resolution crops of the service-manual matrix pages through `.github/workflows/circuit-inspection.yml`, run on GitHub (this environment's proxy blocked the source hosts). Transcribe each cell into `FactoryToneHypotheses.h` with its page, without changing V1 numerics.
Tremolo/chorus "both off/on"	The original panel's two independent switches need one more engine parameter, appended under a new marker, mapped so existing states keep their meaning.
UI gaps	Context menu and MIDI learn for switches, combo boxes and the memory-lid miniatures. Wrap undo-applied values in host begin/end gestures. Feed the voice monitor from engine taps instead of panel settings. Extend the 16-voice meter and pressure ownership to cover the second engine.

3.6 Owner actions (an agent cannot do these)

- **Signing.** Add the secrets listed at the top of `.github/workflows/release.yml` (Developer ID, notarization, Authenticode). Re-run the workflow to get signed installers.
- **Real-DAW sessions.** Live, Logic, Reaper, Bitwig, Cubase and FL Studio, using the checklist in `docs/release/RELEASE_QUALIFICATION.md`.
- **Legal review.** The "CS-80" and "Yamaha" names, the bundle ID, and the engraved YAMAHA nameplate in the editor.
- **AAX.** Requires the Avid SDK and a PACE agreement.

4. Rules the next agent must follow

1. **Legacy never changes.** Run `scripts/check-compat-gates.sh` after every audio change. `legacy default`, `legacy tails` and `v1 circuit numerics` must stay byte-identical.
2. **One V2 re-pin per sound change.** Push the commit that changes circuit-engine audio. Then make a separate `ci: re-pin circuit-engine audio V2 gate …` commit setting `SOURCE_AUDIO_V2` in `dsp-measurements.yml` to that full SHA. Check it with `--strict-v2` and list the change in `docs/STATUS.md`. The current pin is `00ceccf`.
3. **Truth model first.** Commit tolerances before evaluating against them. Mutation-check every new gate. Never weaken, skip or delete a test to get green.
4. **Append-only parameters.** Add new parameters below a new marker in `Params.h` and `createParameterLayout()`, with the next AU version hint (16 or higher). Update the exact parameter count in `tests/test_parity_capture.py`.
5. **PURE CS-80.** Modern features must be inert under `isPureCs80(policy)` and neutral by default.
6. **APVTS state.** Code that copies or replaces it must take `getUndoableState().treeLock()` and call `syncNow()` first.
7. **Honest evidence.** Every law cites a source page or is labelled a declared inference with alternatives. Nothing is presented as a hardware measurement.
8. **Docs.** Update `docs/STATUS.md` (phase table, scorecard, open issues) and `docs/HARDWARE_COVERAGE.md` with dated entries.

9. **Landing changes.** Work on a feature branch and merge to `main` through a PR after CI passes. Refresh the README download links from the new `main` runs.

5. How to verify

```
cmake -S . -B build-core -G Ninja -DCMAKE_BUILD_TYPE=Release -DCS80_BUILD_PLUGIN=OFF -
DCS80_BUILD_TESTS=ON
cmake --build build-core && ctest --test-dir build-core --output-on-failure
scripts/check-compat-gates.sh --strict-v2
cmake --preset linux-release && cmake --build --preset linux-release
ctest --preset linux-release --output-on-failure          # UI smoke, locale, golden images
build-core/cs80_engine_audit --suite all --rates 44100,48000,96000
build-core/cs80_source_factory_benchmark
```

The UI smoke test needs Xvfb, xauth and openbox. JUCE 8.0.11 is fetched by CMake, or can be placed in `deps/juce`. CI then runs `build.yml` (four platforms plus pluginval), `dsp-measurements.yml` (legacy/V1/V2 gates and sanitizers), `screenshots.yml` and, on dispatch, `release.yml`.

6. Where to read more

Topic	Document
Current state, model register, scorecard, open issues	`docs/STATUS.md`
What each workstream did and why	`docs/workstreams/HANDOVER.md` and `docs/workstreams/ws1–ws8-*.md`
The original plan	`docs/IMPROVEMENT_PLAN.md`, `docs/CS80_PLAN_AND_STATUS.pdf`
Per-control coverage	`docs/HARDWARE_COVERAGE.md`, `docs/PANEL_CONFORMANCE.md`
Release and licensing	`docs/release/RELEASE_QUALIFICATION.md`, `LICENSE` (AGPLv3)
Validator crash history	`docs/VALIDATOR_DIAGNOSTICS.md`